

PL/SQL Lab 8 - Detailed Solutions

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1 Task 1 Compute and print bonus based on salary

Problem

Accept an employee number and compute bonus based on salary.

Code

```
1
2 SET SERVEROUTPUT ON;
3 ACCEPT emp_no NUMBER PROMPT 'Enter Employee Number: ';
4 DECLARE
5     v_salary EMPLOYEES.SALARY%TYPE;
6     v_bonus  NUMBER := 0;
7 BEGIN
8     BEGIN
9         SELECT SALARY INTO v_salary
10        FROM EMPLOYEES
11       WHERE EMPLOYEE_ID = &emp_no;
12    EXCEPTION
13        WHEN NO_DATA_FOUND THEN
14            DBMS_OUTPUT.PUT_LINE('Employee not found. ');
15            RETURN;
16    END;
17
18    IF v_salary IS NULL THEN
19        v_bonus := 0;
20    ELSIF v_salary < 1000 THEN
21        v_bonus := v_salary * 0.10;
22    ELSIF v_salary BETWEEN 1000 AND 1500 THEN
23        v_bonus := v_salary * 0.15;
24    ELSE
25        v_bonus := v_salary * 0.20;
26    END IF;
27
28    DBMS_OUTPUT.PUT_LINE('Bonus = ' || v_bonus);
29 END;
30 /
```

2 Task 2 Update salary when commission is NULL

```
1 SET SERVEROUTPUT ON;
2 ACCEPT emp_no NUMBER PROMPT 'Enter Employee ID: ';
3 DECLARE
4     v_comm EMPLOYEES.COMMISSION_PCT%TYPE;
```

```

5      v_sal  EMPLOYEES.SALARY%TYPE;
6 BEGIN
7     BEGIN
8         SELECT COMMISSION_PCT, SALARY
9         INTO v_comm, v_sal
10        FROM EMPLOYEES
11        WHERE EMPLOYEE_ID = &emp_no;
12    EXCEPTION WHEN NO_DATA_FOUND THEN
13        DBMS_OUTPUT.PUT_LINE('Employee not found.');
```

```

14        RETURN;
15    END;
16
17    IF v_comm IS NULL THEN
18        DBMS_OUTPUT.PUT_LINE('Commission is NULL, no update.');
```

```

19    ELSE
20        UPDATE EMPLOYEES
21        SET SALARY = SALARY + (v_comm * SALARY)
22        WHERE EMPLOYEE_ID = &emp_no;
23        DBMS_OUTPUT.PUT_LINE('Salary updated.');
```

```

24    END IF;
25 END;
26 /

```

3 Task 3 Department name for dept 30

```

1 DECLARE
2     v_dname DEPARTMENTS.DEPARTMENT_NAME%TYPE;
3 BEGIN
4     SELECT DEPARTMENT_NAME INTO v_dname
5     FROM DEPARTMENTS
6     WHERE DEPARTMENT_ID = 30;
7
8     DBMS_OUTPUT.PUT_LINE('Department: ' || v_dname);
9 EXCEPTION WHEN NO_DATA_FOUND THEN
10    DBMS_OUTPUT.PUT_LINE('Department not found.');
```

```

11 END;
12 /

```

4 Task 4 Job nature using cursor

```

1 SET SERVEROUTPUT ON;
2 ACCEPT dept_no NUMBER PROMPT 'Enter Dept No: ';
3 DECLARE
4     CURSOR c1 IS

```

```

5      SELECT JOB_ID FROM EMPLOYEES WHERE DEPARTMENT_ID = &dept_no;
6      v_job EMPLOYEES.JOB_ID%TYPE;
7  BEGIN
8      OPEN c1;
9      LOOP
10         FETCH c1 INTO v_job;
11         EXIT WHEN c1%NOTFOUND;
12         DBMS_OUTPUT.PUT_LINE('Job: ' || v_job);
13     END LOOP;
14     CLOSE c1;
15 END;
16 /

```

5 Task 5 Salary of employee in dept 20

```

1 DECLARE
2     v_sal EMPLOYEES.SALARY%TYPE;
3 BEGIN
4     SELECT SALARY INTO v_sal
5     FROM EMPLOYEES
6     WHERE DEPARTMENT_ID = 20 AND ROWNUM = 1;
7
8     DBMS_OUTPUT.PUTLINE('Salary: ' || v_sal);
9 EXCEPTION WHEN NO_DATA_FOUND THEN
10    DBMS_OUTPUT.PUTLINE('No employees in dept 20.');
```

6 Task 6 Procedure increase salary

```

1 CREATE OR REPLACE PROCEDURE increase_salary(emp_id IN NUMBER) AS
2 BEGIN
3     UPDATE EMPLOYEES
4     SET SALARY = SALARY * 1.10
5     WHERE EMPLOYEE_ID = emp_id;
6
7     DBMS_OUTPUT.PUTLINE('Salary updated.');
```

7 Task 7 Procedure add 1000 for salary;5000

```

1 CREATE OR REPLACE PROCEDURE add_bonus(dept_id IN NUMBER) AS
2 BEGIN
3     UPDATE EMPLOYEES
4     SET SALARY = SALARY + 1000
5     WHERE SALARY > 5000 AND DEPARTMENT_ID = dept_id;
6
7     DBMS_OUTPUT.PUTLINE('Bonus added.');
```

8 Task 8 Views

a) Designation + count

```

1 CREATE OR REPLACE VIEW emp_designation_count AS
2 SELECT JOB_ID, COUNT(*) emp_count
3 FROM EMPLOYEES
4 GROUP BY JOB_ID;
```

b) Excluding KING

```

1 CREATE OR REPLACE VIEW emp_except_king AS
2 SELECT E.EMPLOYEE_ID, E.FIRST_NAME, E.DEPARTMENT_ID, D.DEPARTMENT_NAME
3 FROM EMPLOYEES E
4 JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
5 WHERE UPPER(E.FIRST_NAME) != 'KING';
```

c) empno, name, dept, deptname

```

1 CREATE OR REPLACE VIEW emp_details AS
2 SELECT E.EMPLOYEE_ID, E.FIRST_NAME, E.DEPARTMENT_ID, D.DEPARTMENT_NAME
3 FROM EMPLOYEES E
4 JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

9 Task 9 Add two inputs

```

1 SET SERVEROUTPUT ON;
2 ACCEPT a NUMBER PROMPT 'Enter first number: ';
3 ACCEPT b NUMBER PROMPT 'Enter second number: ';
4 DECLARE v NUMBER;
```

```

5 BEGIN
6     v := &a + &b;
7     DBMS_OUTPUT.PUTLINE('Sum: ' || v);
8 END;
9 /

```

10 Task 10 Sum between boundaries

```

1 SET SERVEROUTPUT ON;
2 ACCEPT low NUMBER PROMPT 'Enter lower bound: ';
3 ACCEPT high NUMBER PROMPT 'Enter upper bound: ';
4 DECLARE v_sum NUMBER := 0;
5 BEGIN
6     FOR i IN (&low)..(&high) LOOP
7         v_sum := v_sum + i;
8     END LOOP;
9     DBMS_OUTPUT.PUTLINE('Sum: ' || v_sum);
10 END;
11 /

```

11 Task 11 Name, hiredate, dept

```

1 SET SERVEROUTPUT ON;
2 ACCEPT emp NUMBER PROMPT 'Enter Emp ID: ';
3 DECLARE
4     v_name EMPLOYEES.FIRST_NAME%TYPE;
5     v_hd    EMPLOYEES.HIRE_DATE%TYPE;
6     v_dn    DEPARTMENTS.DEPARTMENT_NAME%TYPE;
7 BEGIN
8     SELECT E.FIRST_NAME, E.HIRE_DATE, D.DEPARTMENT_NAME
9     INTO v_name, v_hd, v_dn
10    FROM EMPLOYEES E
11   JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
12  WHERE E.EMPLOYEE_ID = &emp;
13
14     DBMS_OUTPUT.PUTLINE(v_name || ' - ' || v_hd || ' - ' || v_dn);
15 EXCEPTION WHEN NO_DATA_FOUND THEN
16     DBMS_OUTPUT.PUTLINE('Employee not found. ');
17 END;
18 /

```

12 Task 12 Palindrome

```

1 SET SERVEROUTPUT ON;
2 ACCEPT n NUMBER PROMPT 'Enter number: ';
3 DECLARE
4     v NUMBER := &n;
5     r NUMBER := 0;
6     t NUMBER := &n;
7 BEGIN
8     WHILE t > 0 LOOP
9         r := r * 10 + MOD(t, 10);
10        t := TRUNC(t / 10);
11    END LOOP;
12
13    IF r = v THEN
14        DBMS_OUTPUT.PUTLINE('Palindrome');
15    ELSE
16        DBMS_OUTPUT.PUTLINE('Not palindrome');
17    END IF;
18 END;
19 /

```

13 Task 13 Insert dept + emp

```

1 SET SERVEROUTPUT ON;
2 ACCEPT emp_id NUMBER;
3 ACCEPT fname CHAR;
4 ACCEPT lname CHAR;
5 ACCEPT sal NUMBER;
6 ACCEPT dept NUMBER;
7 ACCEPT dname CHAR;
8
9 DECLARE
10     v_emp    NUMBER := &emp_id;
11     v_fn     VARCHAR2(20) := '&fname';
12     v_ln     VARCHAR2(20) := '&lname';
13     v_sal    NUMBER := &sal;
14     v_dept   NUMBER := &dept;
15     v_dn     VARCHAR2(50) := '&dname';
16 BEGIN
17     BEGIN
18         INSERT INTO DEPARTMENTS VALUES (v_dept, v_dn);
19     EXCEPTION WHEN DUP_VAL_ON_INDEX THEN NULL;
20     END;
21
22     INSERT INTO EMPLOYEES (EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY,
23                           DEPARTMENT_ID)

```

```

23     VALUES (v_emp, v_fn, v_ln, v_sal, v_dept);
24
25     DBMS_OUTPUT.PUTLINE('Inserted.');
```

```

26 END;
27 /
```

14 Task 14 First manager above 90 with salary > 2500

```

1  SET SERVEROUTPUT ON;
2  DECLARE
3      v_mgr EMPLOYEES.MANAGER_ID%TYPE;
4      v_sal EMPLOYEES.SALARY%TYPE;
5      curr NUMBER := 90;
6  BEGIN
7      SELECT MANAGER_ID INTO v_mgr
8      FROM EMPLOYEES
9      WHERE EMPLOYEE_ID = curr;
10
11     LOOP
12         IF v_mgr IS NULL THEN
13             DBMS_OUTPUT.PUTLINE('Reached top of chain.');
```

```

14             EXIT;
15         END IF;
16
17         SELECT SALARY INTO v_sal
18         FROM EMPLOYEES WHERE EMPLOYEE_ID = v_mgr;
19
20         IF v_sal > 2500 THEN
21             DBMS_OUTPUT.PUTLINE('Found: ' || v_mgr || ' salary ' ||
22                                 v_sal);
23             EXIT;
24         END IF;
25
26         SELECT MANAGER_ID INTO v_mgr
27         FROM EMPLOYEES WHERE EMPLOYEE_ID = v_mgr;
28     END LOOP;
29 END;
/
```

15 Task 15 Sum of first 100 numbers

```

1  SET SERVEROUTPUT ON;
2  DECLARE v NUMBER := 0;
3  BEGIN
```



```
4      FOR i IN 1..100 LOOP
5          v := v + i;
6      END LOOP;
7      DBMS_OUTPUT.PUTLINE('Sum = ' || v);
8  END;
9  /
```